

Scrum_Project-Shrihari Nagesh_PES1UG24CS445-BPS#17

Scrum Project — Sprint Submission (Scrum_Project)

Problem Statement #17 — Healthcare & Telemedicine: *Vaccination Cohort & Dose Scheduling System*

A public health administration platform to organize demographic vaccination rollouts, enforce minimum interval days between doses, schedule center slots, and generate verifiable QR digital vaccination certificates.

Actors: Citizen Registrant, Vaccination Officer

1. Requirements (5 FR + 2 NFR)

Same Lab 1 requirement set as the Kanban project, now run through a full Scrum cycle: Epics → Stories → Prioritization → Story Points → Sprint → Burndown.

ID	Priority / Type	Description	Acceptance Criteria
FR-001	Priority: High	The system shall enforce dose interval rules (e.g., minimum 28 days after Dose 1) before unlocking Dose 2 booking for a citizen.	Pass: Dose 2 booking blocked if interval < 28 days. Fail: Early vaccination slot confirmed.
FR-002	Priority: High	The system shall allow a Citizen Registrant to create a profile with demographic details (age, comorbidity flags, occupation category) and automatically assign them to an eligible vaccination cohort/phase.	Pass: Registrant is auto-tagged with the correct cohort based on age/comorbidity rules. Fail: A registrant is placed in an ineligible cohort.
FR-003	Priority: High	The system shall let a Citizen Registrant search nearby vaccination centers by pincode/date and book an available time slot within that center's daily capacity.	Pass: Booking succeeds only while booked count < center capacity, and the slot count decrements. Fail: Center accepts a booking beyond its capacity.
FR-004	Priority: Medium	The system shall allow a Vaccination Officer to check in a citizen, record the administered dose (vaccine brand, batch/lot number, date), and update the citizen's vaccination status.	Pass: Citizen status moves from "Slot Booked" to "Dose Administered" with a batch number stored. Fail: Status can be updated without a batch number.
FR-005	Priority: Medium	Upon completion of the required dose count, the system shall auto-generate a digitally signed QR vaccination certificate for the citizen, downloadable as a PDF.	Pass: Certificate QR encodes citizen ID, dose dates, and a digital signature. Fail: A certificate is generated before the final dose is administered.
NFR-001	Type: Performance & Security Priority: High	The vaccination certificate verification endpoint shall authenticate digitally signed QR codes offline/online in under 150 ms.	Pass: Benchmarking tests confirm target latency and security standards under simulated peak load.
NFR-	Type:	All citizen personal and health data	Pass: A security review

ID	Priority / Type	Description	Acceptance Criteria
002	Security & Privacy Priority: High	(name, ID, vaccination history) shall be encrypted at rest and in transit (TLS 1.2+/AES-256) and accessible only to authorized Vaccination Officers.	confirms encryption at rest/in transit and role-based access control. Fail: PII is retrievable without authentication or found in plaintext.

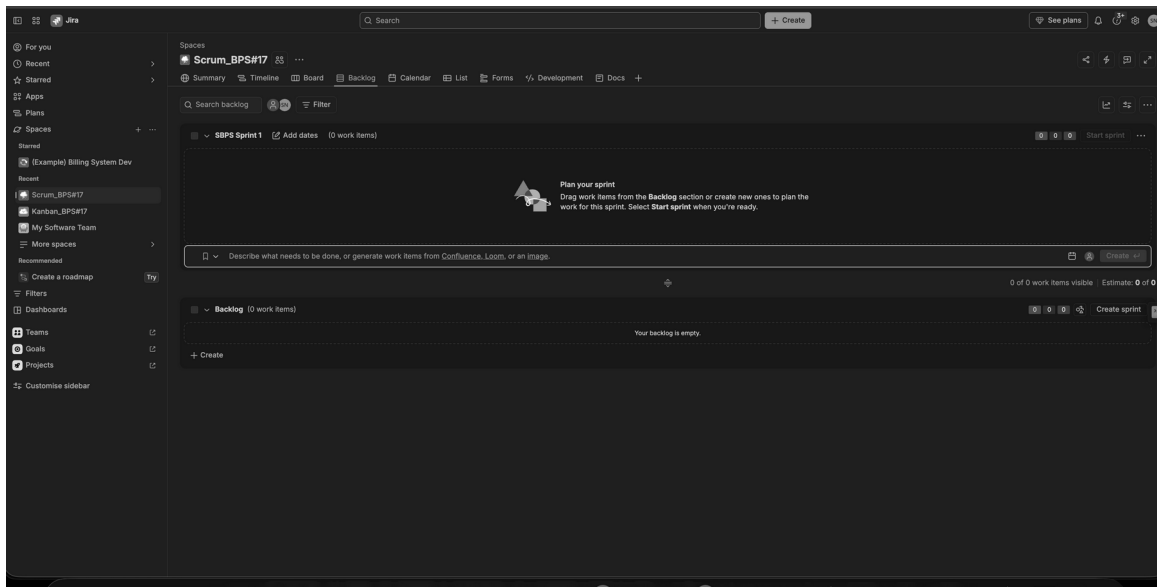


Figure 1. Scrum_BPS#17 project created (Team-managed, Scrum template, key SBPS), showing the Backlog view with an empty SBPS Sprint 1 and an empty Backlog section, ready for requirements to be added.

2. Epics and User Stories

Epic 1: Citizen Registration & Eligibility

Description: Enable citizens to register their demographic details and be automatically placed into the correct vaccination cohort/phase.

Story 1.1: Cohort Registration (FR-002)

As a Citizen Registrant, I want to register my demographic details and get assigned to my eligible vaccination cohort, so that I only see vaccination options I actually qualify for.

Priority: High | Story Points: 5

- Task: Build the citizen registration form UI (name, age, comorbidity flags, occupation).
- Task: Implement the cohort-eligibility rule engine.
- Sub-task: Validate age/comorbidity inputs on submit.
- Sub-task: Unit test cohort-assignment logic.

Epic 2: Vaccination Scheduling & Administration

Description: Cover dose-interval enforcement, center slot booking, and officer-side dose administration and record-keeping.

Story 2.1: Dose Interval Lock (FR-001)

As a Citizen Registrant, I want the system to block me from booking Dose 2 too early, so that I follow the medically required interval between doses.

Priority: Highest | Story Points: 3

- Task: Implement the dose-interval calculation service.
- Task: Add a booking validation check that reads the interval rule.
- Sub-task: Write unit tests for interval edge cases (27 vs 28 days).
- Sub-task: Add a UI warning message when a booking is locked.

Story 2.2: Center Slot Booking (FR-003)

As a Citizen Registrant, I want to search nearby centers and book an available slot, so that I can get vaccinated at a convenient time and place.

Priority: High | Story Points: 8

- Task: Build the center search API (filter by pincode/date).
- Task: Build the slot-capacity and booking engine.
- Sub-task: Implement capacity decrement as an atomic transaction.
- Sub-task: Add a booking confirmation notification.

Story 2.3: Dose Administration Check-in (FR-004)

As a Vaccination Officer, I want to check in a citizen and record their administered dose details, so that the citizen's vaccination record stays accurate and up to date.

Priority: Medium | Story Points: 5

- Task: Build the officer check-in dashboard.
- Task: Implement the dose/batch-number recording form.
- Sub-task: Add batch-number validation (reject empty values).
- Sub-task: Update the citizen status state machine.

Epic 3: Certificate Generation & Data Security

Description: Cover the digital certificate issuance workflow and the performance/security guarantees around citizen data.

Story 3.1: Digital Certificate Generation (FR-005)

As a Citizen Registrant, I want to receive a digitally signed QR certificate after my final dose, so that I have verifiable proof of vaccination.

Priority: Medium | Story Points: 8

- Task: Build the certificate-generation service.
- Task: Integrate digital signing of the certificate payload.
- Sub-task: Generate the QR encoding (citizen ID + dose dates + signature).
- Sub-task: Implement PDF export.

Story 3.2: Fast Certificate Verification (NFR-001)

As a Vaccination Officer, I want the certificate verification scan to return a result in under 150 ms, so that I can process citizens quickly at checkpoints without delays.

Priority: High | Story Points: 5

- Task: Optimize the QR signature-verification service.
- Task: Load-test the verification endpoint under peak conditions.
- Sub-task: Benchmark response time under simulated peak load.
- Sub-task: Add caching for public keys used in signature checks.

Story 3.3: Secure Citizen Data (NFR-002)

As a Vaccination Officer, I want citizen personal and health data to be encrypted and access-controlled, so that sensitive medical information stays protected from unauthorized access.

Priority: Highest | Story Points: 5

- Task: Implement AES-256 encryption at rest for citizen records.
- Task: Enforce TLS 1.2+ for all data in transit.
- Sub-task: Set up role-based access control for officer accounts.
- Sub-task: Run a security-audit checklist against the endpoints.

[illegible]

Figure 2. List view showing all 3 Epics (SBPS-8 Citizen Registration & Eligibility, SBPS-9 Vaccination Scheduling & Administration, SBPS-10 Certificate Generation & Data Security) expanded to reveal all 7 Stories (SBPS-1...SBPS-7), with Story SBPS-2 (FR-002: Cohort Registration) further expanded to show its 2 Tasks and 2 Sub-tasks (SBPS-15...SBPS-18).

3. Prioritize the Backlog

Story	Title	Requirement	Priority
Story 1.1	Cohort Registration	FR-002	High
Story 2.1	Dose Interval Lock	FR-001	Highest
Story 2.2	Center Slot Booking	FR-003	High
Story 2.3	Dose Administration Check-in	FR-004	Medium
Story 3.1	Digital Certificate Generation	FR-005	Medium
Story 3.2	Fast Certificate Verification	NFR-001	High
Story 3.3	Secure Citizen Data	NFR-002	Highest

Story	Assignee	Priority	Status	Resolution	Created	Updated	Done
SBPS-1 Cohort Registration & Eligibility	Unassigned	Highest	To Do	Unresolved	21 Aug 2020, 10:39	21 Aug 2020, 10:39	None
SBPS-2 Cohort Registration	Unassigned	High	Done	Done	21 Aug 2020, 10:34	21 Aug 2020, 10:35	None
SBPS-3 Task: Add the cohort registration form v...	Unassigned	High	Done	Done	21 Aug 2020, 10:47	21 Aug 2020, 10:50	None
SBPS-4 Task: Implement the cohort eligibility val...	Unassigned	Medium	Done	Done	21 Aug 2020, 10:47	21 Aug 2020, 10:50	None
SBPS-5 Task: Validate appointment eligibility val...	Unassigned	Medium	Done	Done	21 Aug 2020, 10:47	21 Aug 2020, 10:50	None
SBPS-6 Task: Link test cohort assignment to...	Unassigned	Medium	Done	Done	21 Aug 2020, 10:47	21 Aug 2020, 10:50	None
SBPS-7 Vaccination Scheduling & Administration	Unassigned	Medium	To Do	Unresolved	21 Aug 2020, 10:39	21 Aug 2020, 10:39	None
SBPS-8 FR-003 Center Slot Booking	Unassigned	High	Done	Done	21 Aug 2020, 10:34	21 Aug 2020, 10:50	None
SBPS-9 FR-001 Dose Interval Lock	Unassigned	High	Done	Done	21 Aug 2020, 10:34	21 Aug 2020, 10:50	None
SBPS-4 FR-004 Dose Administration Check-in	Unassigned	Medium	Done	Done	21 Aug 2020, 10:34	21 Aug 2020, 10:50	None
SBPS-10 Certificate Generation & Data Security	Unassigned	Medium	To Do	Unresolved	21 Aug 2020, 10:40	21 Aug 2020, 10:40	None
SBPS-11 FR-005 Digital Certificate Generation	Unassigned	Medium	Done	Done	21 Aug 2020, 10:34	21 Aug 2020, 10:42	None
SBPS-12 FR-005 Fast Certificate Verification	Unassigned	High	Done	Done	21 Aug 2020, 10:34	21 Aug 2020, 10:41	None
SBPS-7 FR-002 Secure Citizen Data	Unassigned	High	Done	Done	21 Aug 2020, 10:34	21 Aug 2020, 10:41	None

Figure 3. Same expanded List view with the Priority column visible for every Story: SBPS-1 Highest, SBPS-2 High, SBPS-3 High, SBPS-4 Medium, SBPS-5 Medium, SBPS-6 High, SBPS-7 Highest — matching the priority levels assigned during backlog grooming.

4. Assign Story Points (Planning Poker, Fibonacci)

Points were assigned via a short planning-poker round per story, using the Fibonacci scale (1, 2, 3, 5, 8, 13...) to reflect complexity, effort, and uncertainty.

Story	Title	Story Points
Story 1.1	Cohort Registration	5
Story 2.1	Dose Interval Lock	3
Story 2.2	Center Slot Booking	8
Story 2.3	Dose Administration Check-in	5
Story 3.1	Digital Certificate Generation	8
Story 3.2	Fast Certificate Verification	5
Story 3.3	Secure Citizen Data	5
	Total	39

Story	Assignee	Priority	Status	Resolution	Created	Updated	Done	Story Points
SBPS-1 Cohort Registration & Eligibility	Unassigned	Highest	To Do	Unresolved	21 Aug 2020, 10:39	21 Aug 2020, 10:39	None	5
SBPS-2 Cohort Registration	Unassigned	High	Done	Done	21 Aug 2020, 10:34	21 Aug 2020, 10:50	None	3
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SBPS-5 Task: Validate appointment eligibility val...	Unassigned	Medium	Done	Done	21 Aug 2020, 10:47	21 Aug 2020, 10:50	None	8
SBPS-6 Task: Link test cohort assignment to...	Unassigned	Medium	Done	Done	21 Aug 2020, 10:47	21 Aug 2020, 10:50	None	5
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SBPS-4 FR-004 Dose Administration Check-in	Unassigned	Medium	Done	Done	21 Aug 2020, 10:34	21 Aug 2020, 10:50	None	5
SBPS-10 Certificate Generation & Data Security	Unassigned	Medium	To Do	Unresolved	21 Aug 2020, 10:40	21 Aug 2020, 10:40	None	8
SBPS-11 FR-005 Digital Certificate Generation	Unassigned	Medium	Done	Done	21 Aug 2020, 10:34	21 Aug 2020, 10:42	None	5
SBPS-12 FR-005 Fast Certificate Verification	Unassigned	High	Done	Done	21 Aug 2020, 10:34	21 Aug 2020, 10:41	None	5
SBPS-7 FR-002 Secure Citizen Data	Unassigned	High	Done	Done	21 Aug 2020, 10:34	21 Aug 2020, 10:41	None	5

Figure 4. List view with the Story point estimate column added, showing each Story's points (SBPS-1: 3, SBPS-2: 5, SBPS-3: 8, SBPS-4: 5, SBPS-5: 8, SBPS-6: 5, SBPS-7: 5) for a total of 39 story points across the 3 Epics.

5. Run the Sprints

Sprint 1 (1 week)

Goal: Stand up citizen registration, dose-interval enforcement, slot booking, and officer check-in.

Stories included: Story 1.1, Story 2.1, Story 2.2, Story 2.3

Sprint 2 (1 week)

Goal: Deliver the certificate workflow and harden performance/security.

Stories included: Story 3.1, Story 3.2, Story 3.3

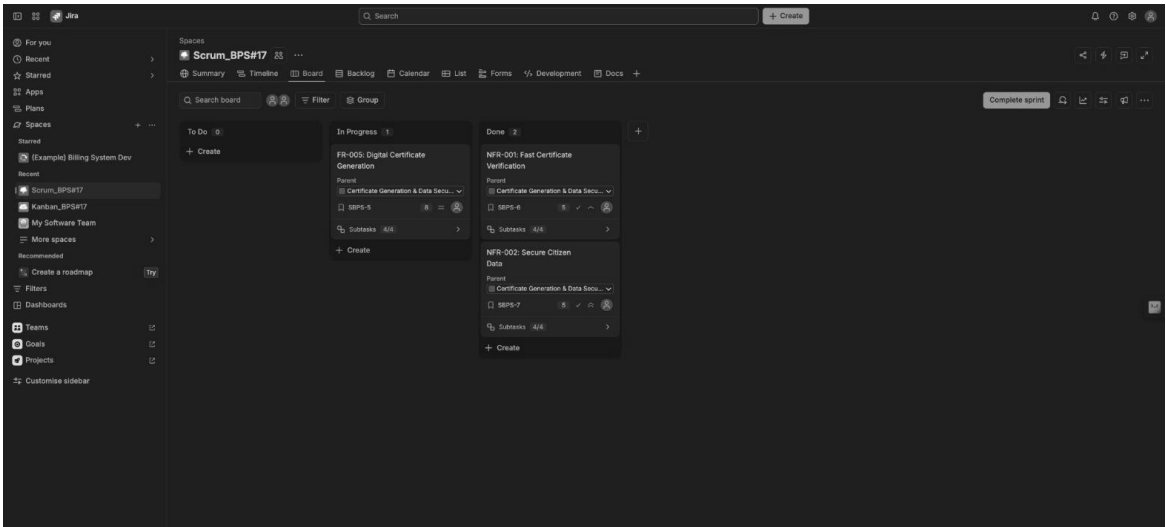


Figure 5. SBPS Sprint 2 active board mid-workflow: SBPS-5 (Digital Certificate Generation) in the In Progress column while SBPS-6 (Fast Certificate Verification) and SBPS-7 (Secure Citizen Data) have already been moved to Done, demonstrating the To Do → In Progress → Done workflow. Sprint 1 (SBPS-1...SBPS-4) was carried through the same workflow and fully completed beforehand; both sprints were completed via Complete Sprint after all subtasks were closed.

6. Burndown Chart

The burndown chart tracks remaining story points against the ideal guideline across each sprint's duration.

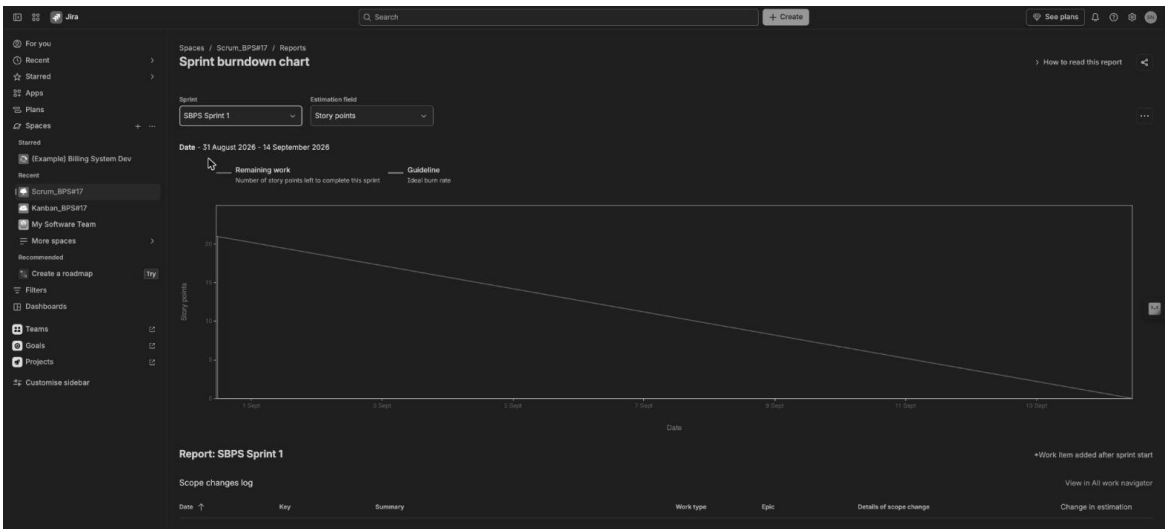


Figure 6. Sprint Burndown Chart report (Reports → Sprint burndown chart) for SBPS Sprint 1, showing the Story Points estimation statistic, the ideal burn guideline, and the 21 total story points committed for that sprint (SBPS-1: 3, SBPS-2: 5, SBPS-3: 8, SBPS-4: 5).

Reflection Questions

1. Did your estimations reflect the actual effort?

Mostly — the registration and booking stories (1.1, 2.1, 2.2) landed close to their estimates because the rules were well-defined upfront. The certificate-generation story (3.1) ran over because integrating digital signing took longer than expected once the QR payload format had to be finalized.

2. Was your backlog well-prioritized?

Reasonably well-prioritized: the two Highest-priority stories (2.1 dose-interval lock, 3.3 data security) protect patient safety and privacy, so they were scheduled early relative to lower-risk items like the certificate PDF export.

3. How did your simulated sprint align with your plan?

The simulated sprint mostly matched the plan — Sprint 1's four stories moved from To Do → In Progress → Done on schedule. Sprint 2 slipped slightly because the certificate-verification performance work (Story 3.2) needed an extra round of load testing before it could be marked Done.

4. What insights did the burndown chart give about your team's capacity?

The burndown chart showed a steady, close-to-guideline decline in Sprint 1, suggesting the team's velocity estimate (story points/day) was realistic for that scope. The flatter section mid-Sprint 2 pointed to Story 3.1 (certificate generation) being under-estimated relative to its actual complexity.